tf.split Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/split

> *Splits a tensor value into a list of sub tensors.*

Function Signatures

```
tf.split( value, num_or_size_splits, axis=0, num=None,
name='split' )
```

Code Examples

```
tf.split(
value, num_or_size_splits, axis=0, num=None, name='split'
)
```

```
tf.split(
value, num_or_size_splits, axis=0, num=None, name='split'
)
```

```
x = tf.Variable(tf.random.uniform([5, 30], -1, 1))
# Split `x` into 3 tensors along dimension 1
s0, s1, s2 = tf.split(x, num_or_size_splits=3, axis=1)
tf.shape(s0).numpy()
array([ 5, 10], dtype=int32)
# Split `x` into 3 tensors with sizes [4, 15, 11] along dimension 1
split0, split1, split2 = tf.split(x, [4, 15, 11], 1)
tf.shape(split0).numpy()
array([5, 4], dtype=int32)
tf.shape(split1).numpy()
array([ 5, 15], dtype=int32)
tf.shape(split2).numpy()
array([ 5, 11], dtype=int32)
```

```
x = tf.Variable(tf.random.uniform([5, 30], -1, 1))
```

```
# Split `x` into 3 tensors along dimension 1
```

```
s0, s1, s2 = tf.split(x, num_or_size_splits=3, axis=1)
```

```
tf.shape(s0).numpy()
```

Documentation

Splits a tensor value into a list of sub tensors.

View aliases

Compat aliases for migration

See
Migration guide for

more details.

tf.compat.v1.split

Used in the notebooks

• Distributed training with Core APIs and DTensor

- Understanding masking & padding
- Using DTensors with Keras
- Distributed training with DTensors
- Convolutional Variational Autoencoder
- MoViNet for streaming action recognition
- Bayesian Modeling with Joint Distribution

See also tf.unstack.

If num_or_size_splits is an int, then it splits value along the
dimension axis into num_or_size_splits smaller tensors. This requires that
value.shape[axis] is divisible by num_or_size_splits.

If num_or_size_splits is a 1-D Tensor (or list), then value is split into
len(num_or_size_splits) elements. The shape of the i-th
element has the same size as the value except along dimension axis where
the size is num_or_size_splits[i].

For example:

Returns

Raises